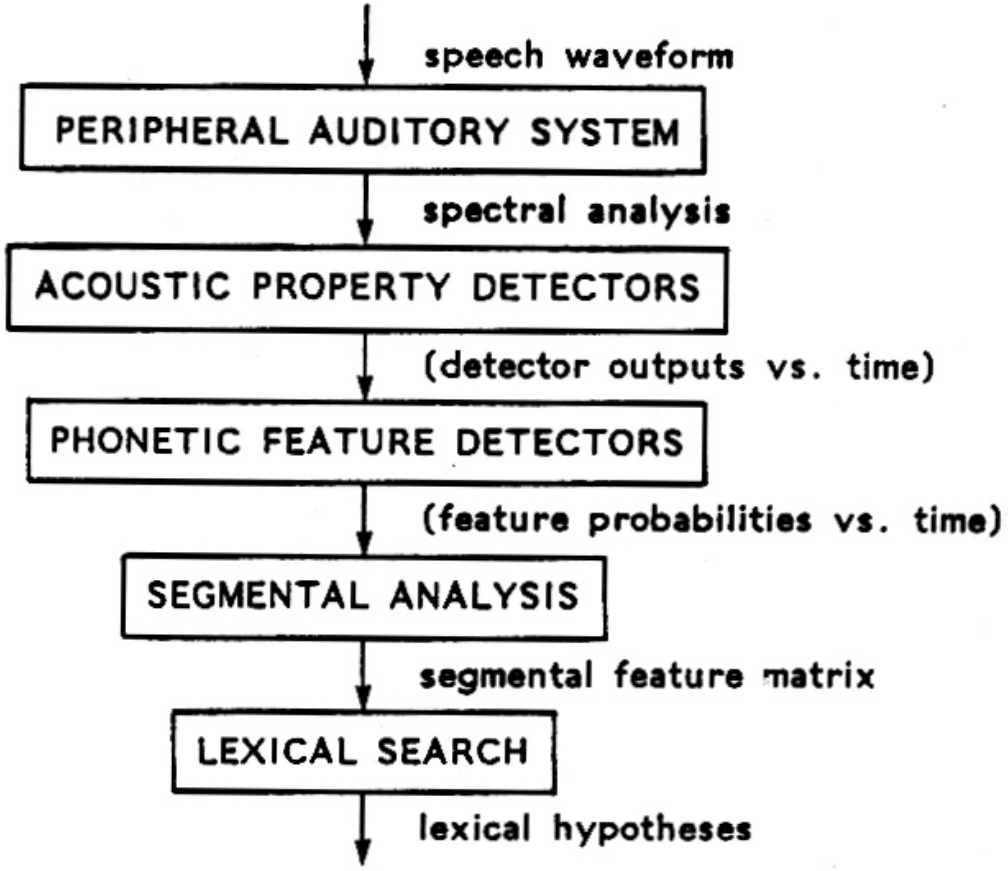
Speech Perception II: Words

A very old and very general model of speech perception



Klatt (1979, J Phon)

On Monday, we discussed categorical perception.

The idea is that somewhere in the journey from the analog signal that comes in through the cochlea, and the activation of phonemes (segmental analysis), information is *lost*.

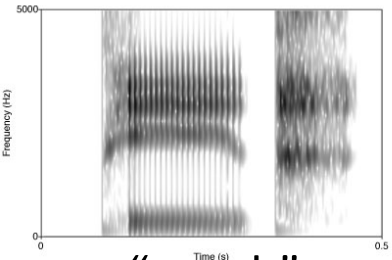
This *loss of information is helpful*, because it allows us to make contact with higher levels of representation.

For example, we might assume that *in order to identify words*, we need to first identify their constituent phonemes.

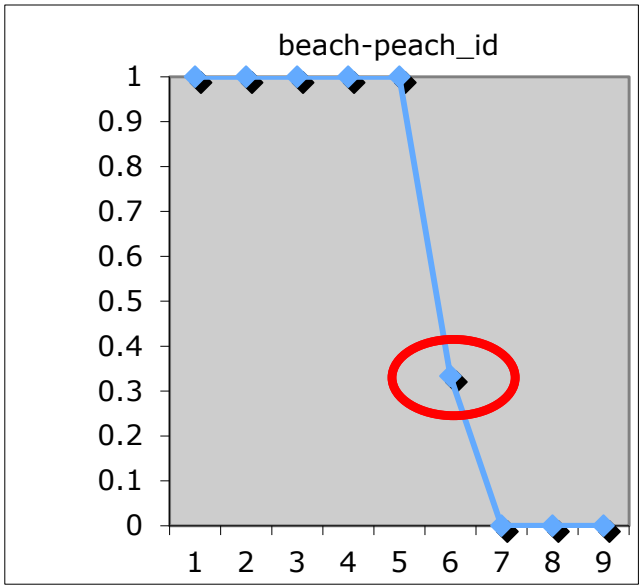
Today we'll talk about two problems for this simple view:

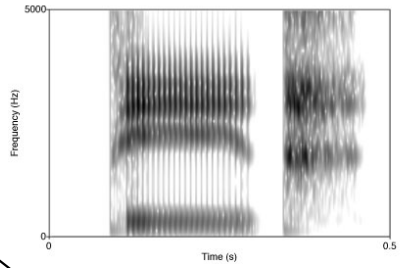
1. **Cascaded processing, and**
2. **Feedback**

A (familiar) voicing continuum.

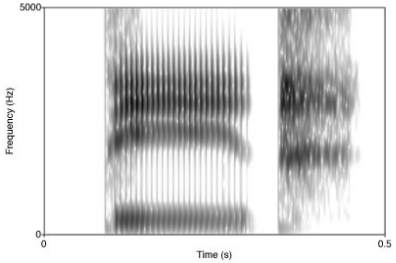


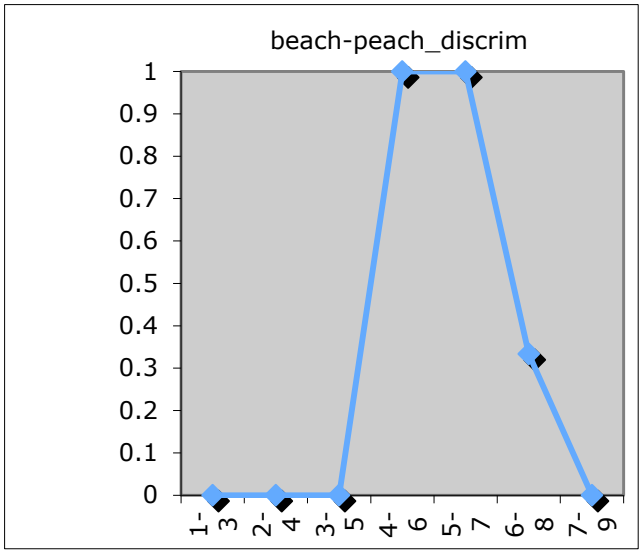
"peach"



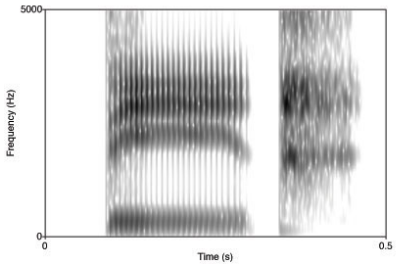


shorter voice onset time

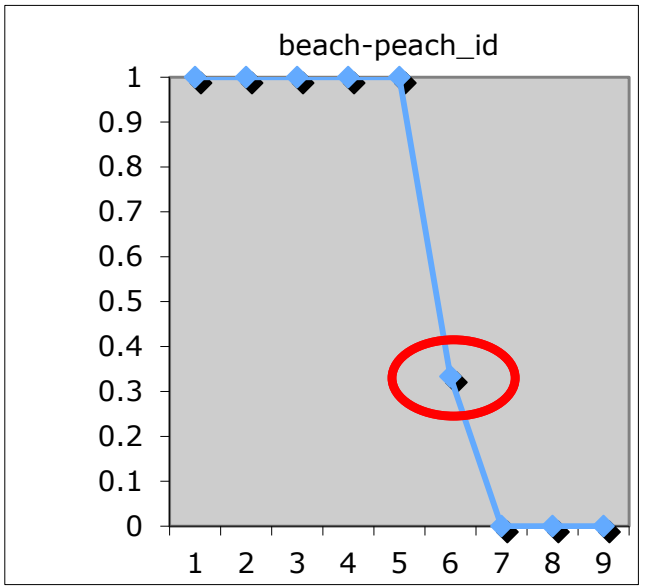




"beach"

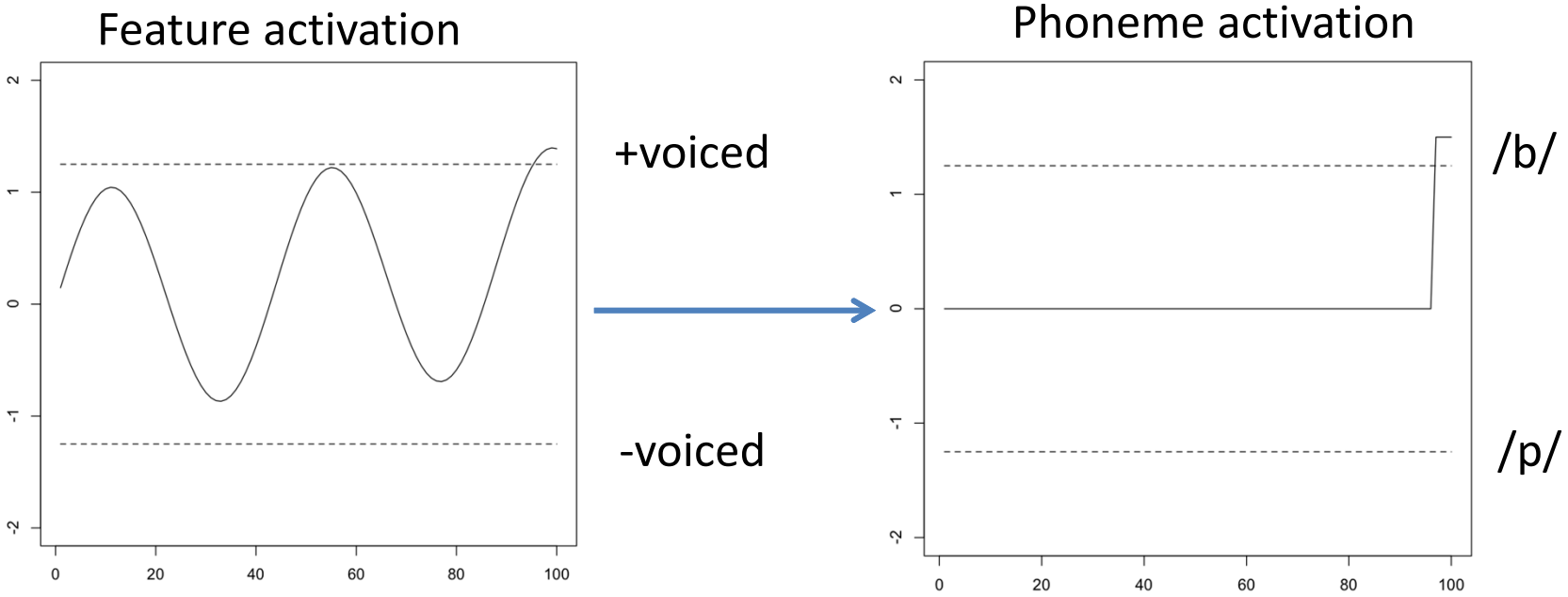


How does categorical perception happen?

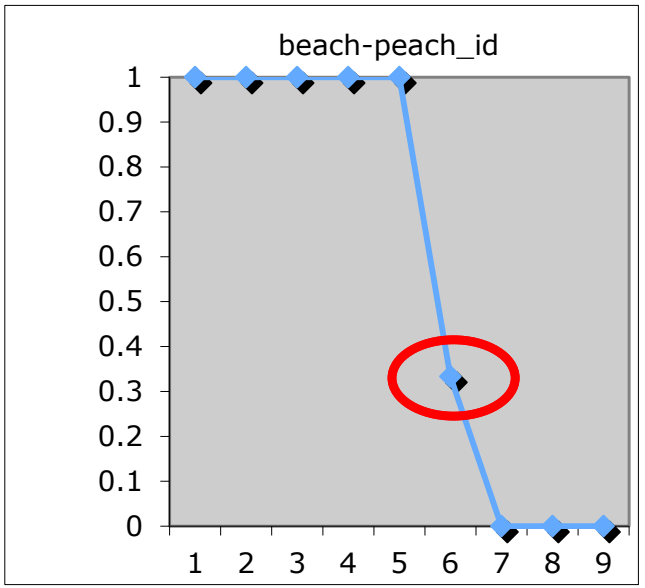


A threshold model: activity is only passed forward from one level to the next when an activity threshold is crossed.

In this kind of model, categorical perception arises at phoneme identification.

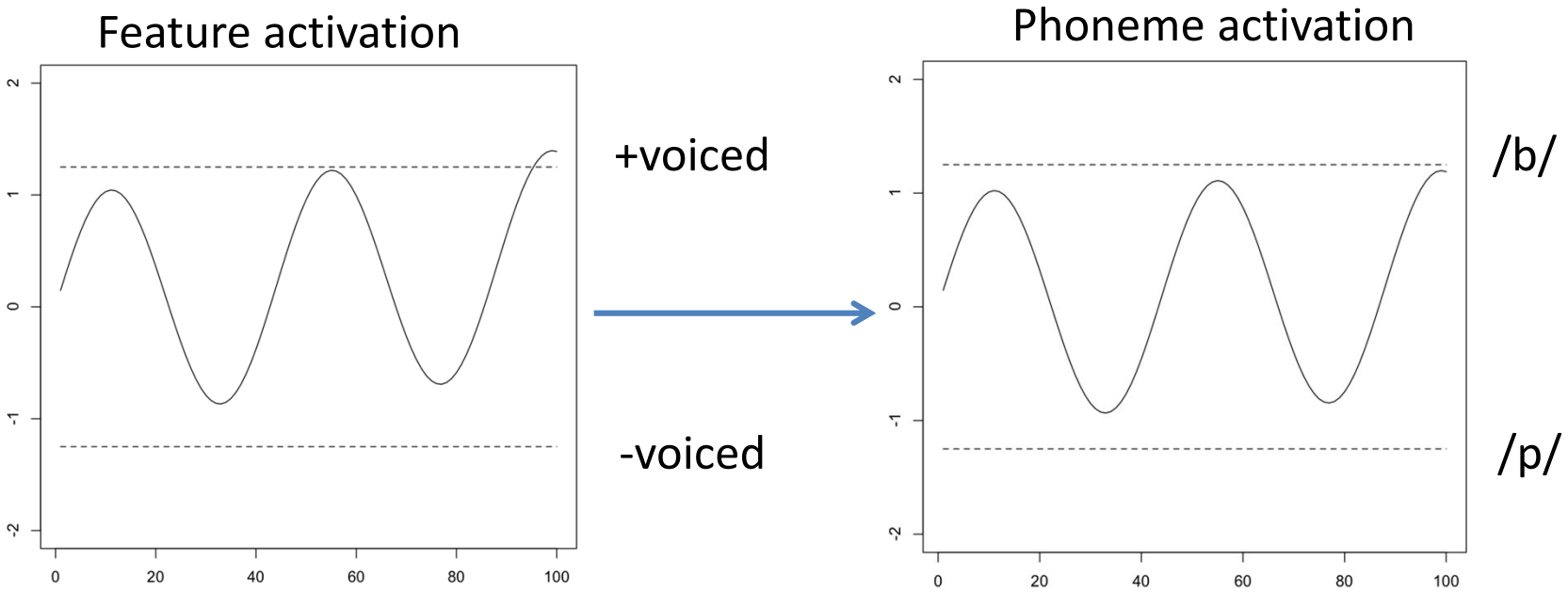


How does categorical perception happen?



A cascaded model: activity is passed forward continuously over time from one level to the next.

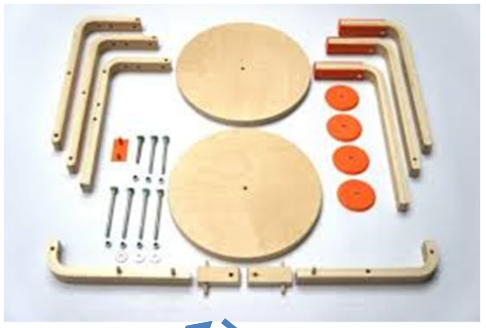
In this kind of model, categorical perception might happen later on, or might even be an artifact of the task structure.



A metaphor for thresholded activity: Making furniture.















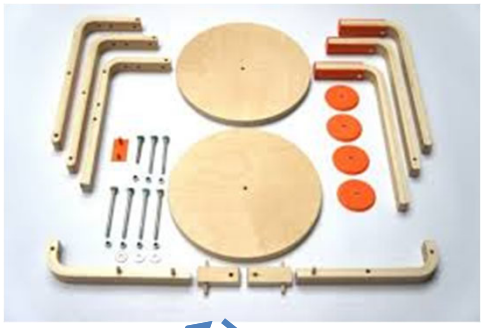
Lumber Mill
Input: trees
Output: lumber

Furniture factory
Input: lumber
Output: unassembled furniture

IKEA customer
Input: unassembled furniture
Output: furniture

A metaphor for thresholded activity: Making furniture.



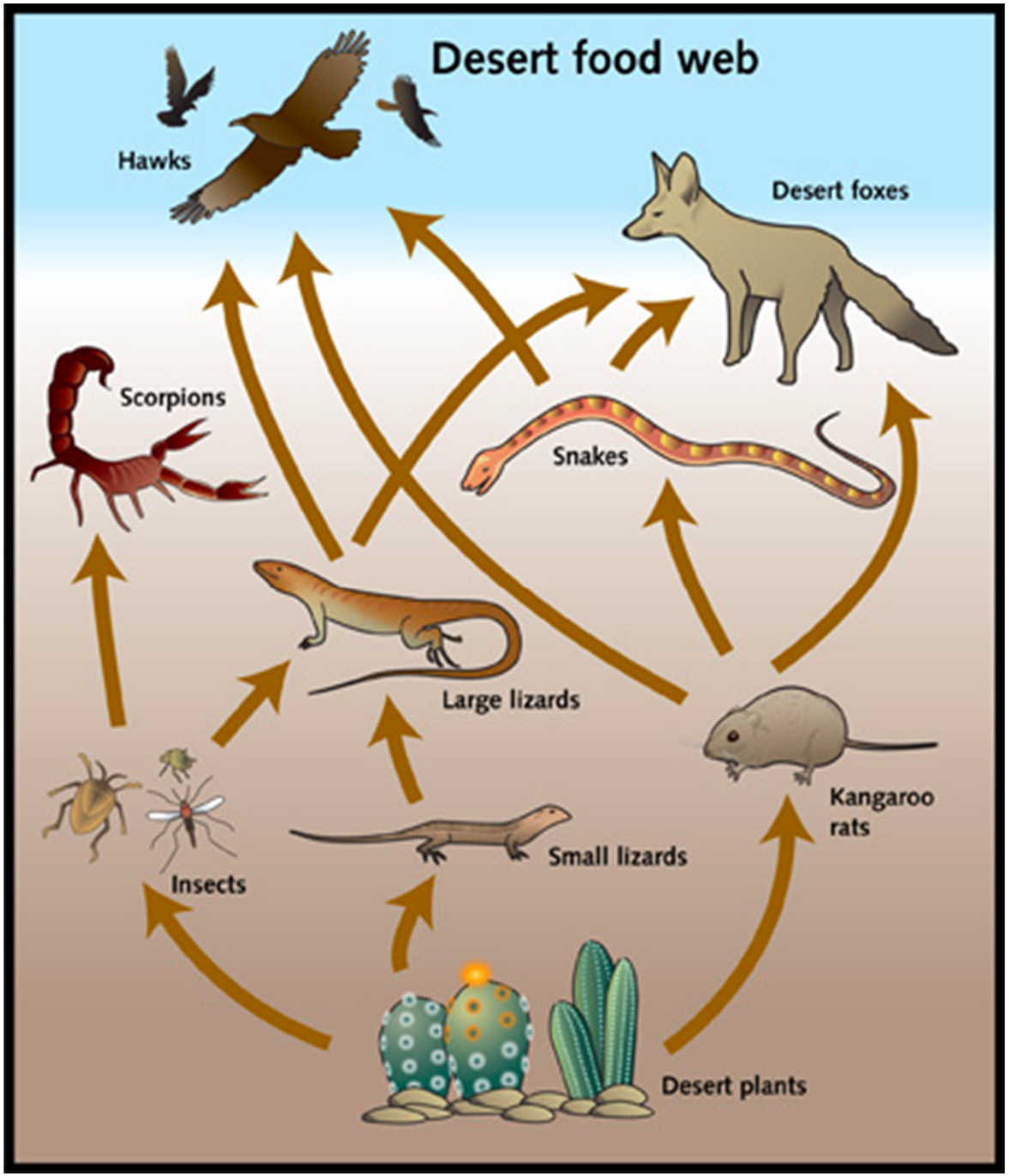







The point is that each “layer” of the system can only take a specific input that’s been transformed *completely* by the previous layer. You have to wait for the shipment of lumber before you can start manufacturing, etc.

(Many computer programs function this way.)



A metaphor for "cascaded" processing:
A "food web."

"levels" continuously interact with one another, so that a drought reducing the number of plants will immediately have an impact on the health of kangaroo rats, which impacts the snakes, etc.

(Further, there are clear feedback loops in here…what impact does adding more hawks have on populations of snakes, lizards and rats?)

So, we want to know, is phoneme identification for word recognition more like thresholded (lumber mill) or cascaded (food web) processing?

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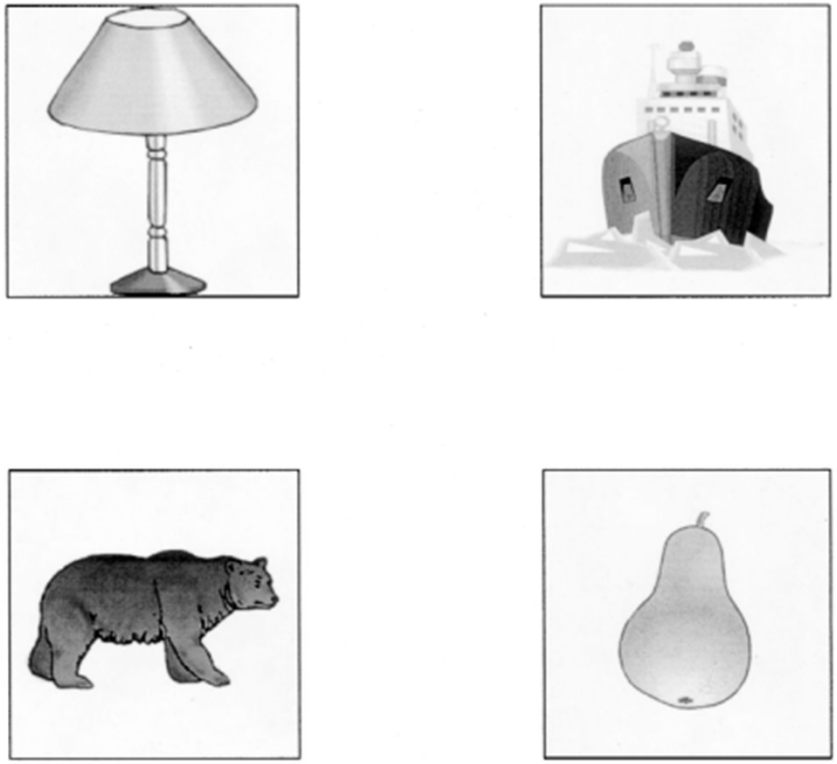


Fig. 3. A sample display. Each display contains a target (e.g., the "pear"), a competitor (e.g., the "bear"), and two filler items ("lamp" and "ship").

McMurray et al. tested identification of a /b/-/p/ continuum in a "visual world" display for eye-tracking.

McMurray, B., Tanenhaus, M., Aslin, R. and Spivey, M. (2003). Probabilistic constraint satisfaction at the lexical/phonetic interface: Evidence for gradient effects of within-category VOT on lexical access. *Journal of Psycholinguistic Research, 32*, 77-97.

http://www2.psychology.uiowa.edu/faculty/mcmurray/publications/mcmurray-jpr.pdf

So, we want to know, is phoneme identification for word recognition more like thresholded (lumber mill) or cascaded (food web) processing?

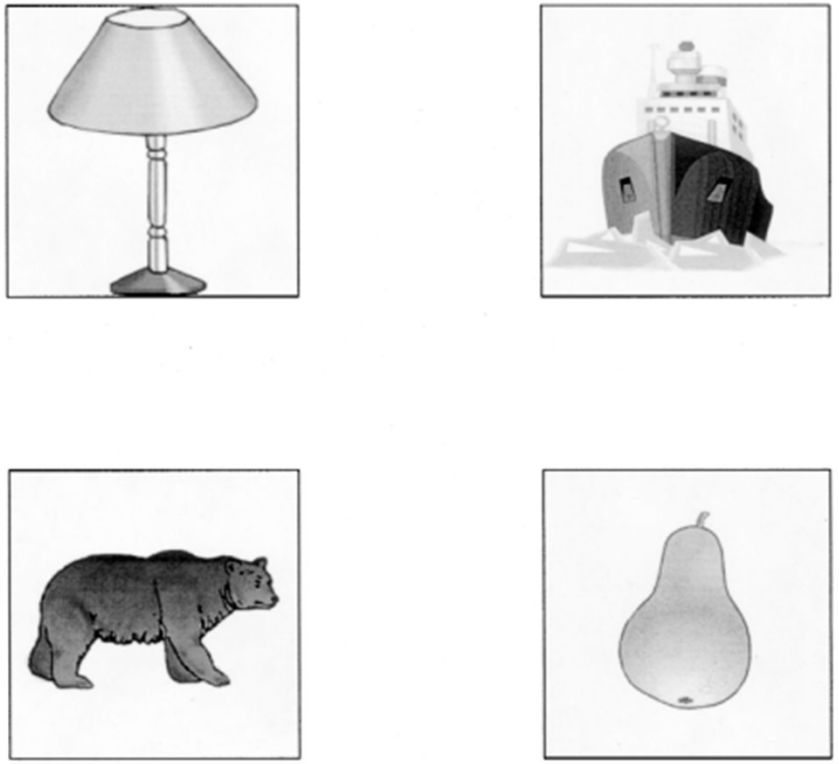


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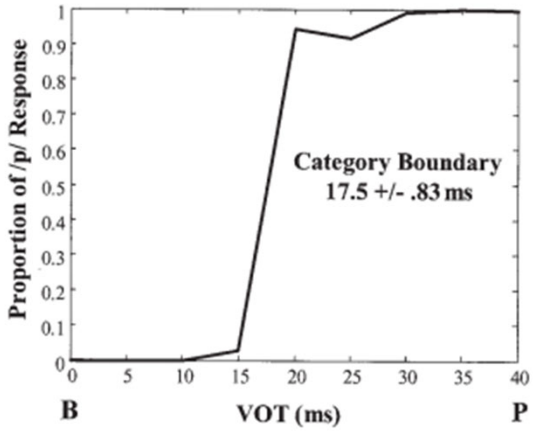


Fig. 1. Identification function (from mouse responses) as a function of VOT.

Participants used the mouse to select the picture most consistent with what they heard, e.g., a "bear" or a "pear," just like in the experiment you did at home.

A typical, step-like, identification function was observed.

Eye-movements reveal that ambiguity is *gradient,* i.e., there are more looks to the competitor close to the boundary.

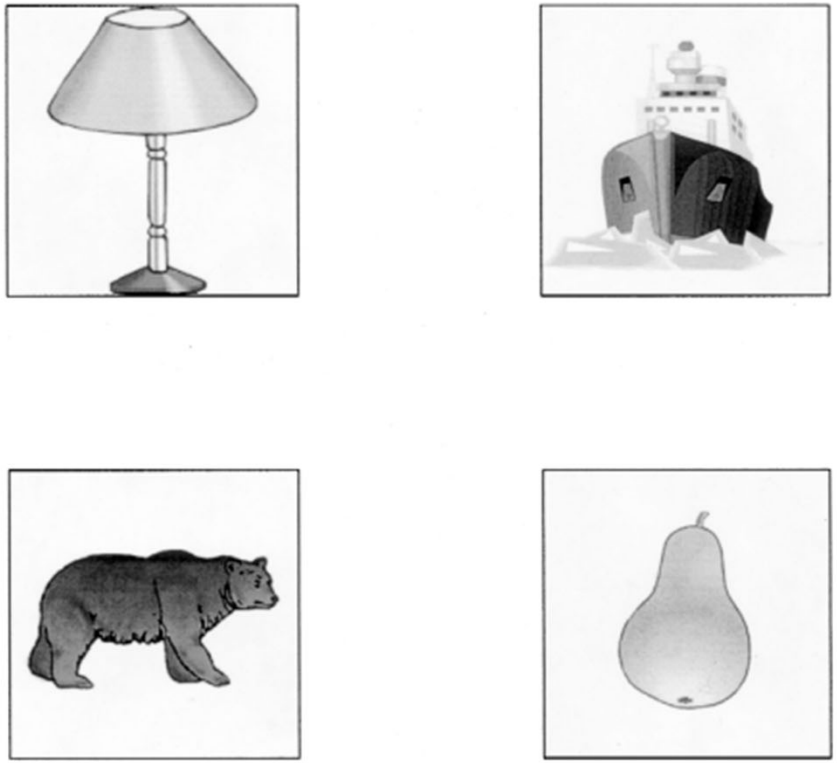


Fig. 3. A sample display. Each display contains a target (e.g., the "pear"), a competitor (e.g., the "bear"), and two filler items ("lamp" and "ship").

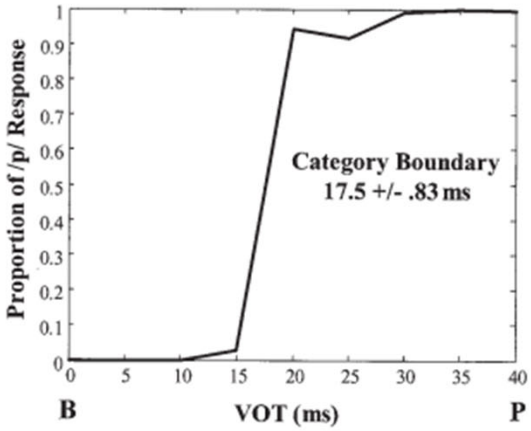


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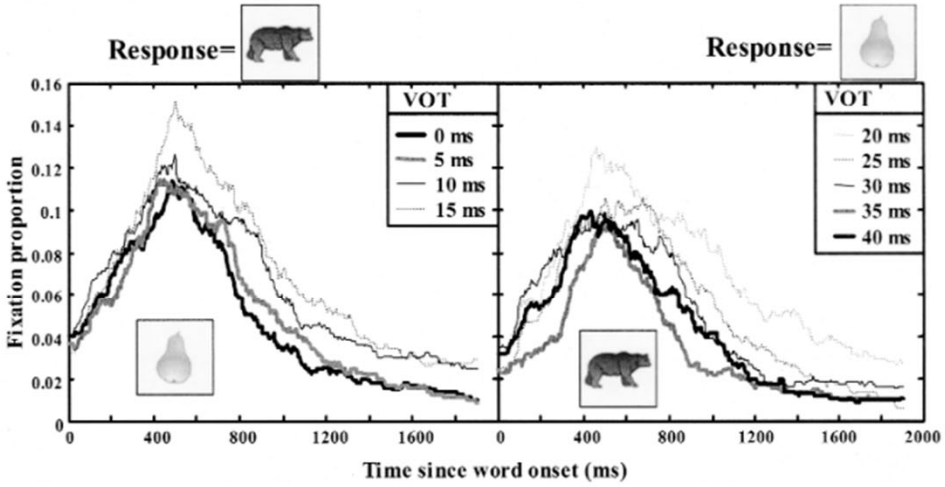


Fig. 7. Effect of VOT and time on fixations to the competitor. A clear gradient effect of VOT can be seen. Importantly, the effect of VOT is primarily on the duration of activation.

The effect is a lot clearer viewed this way…

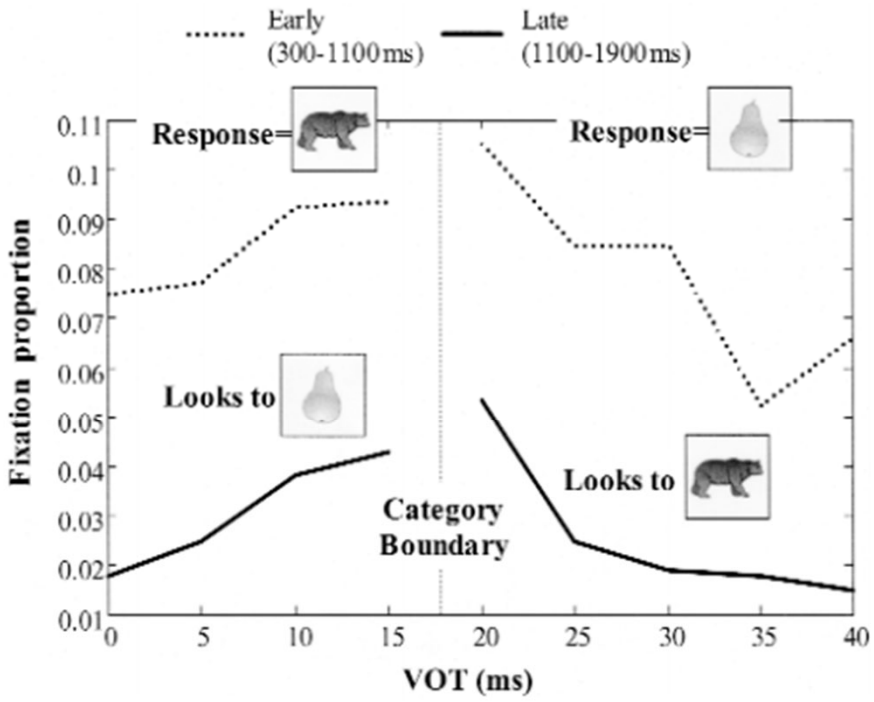
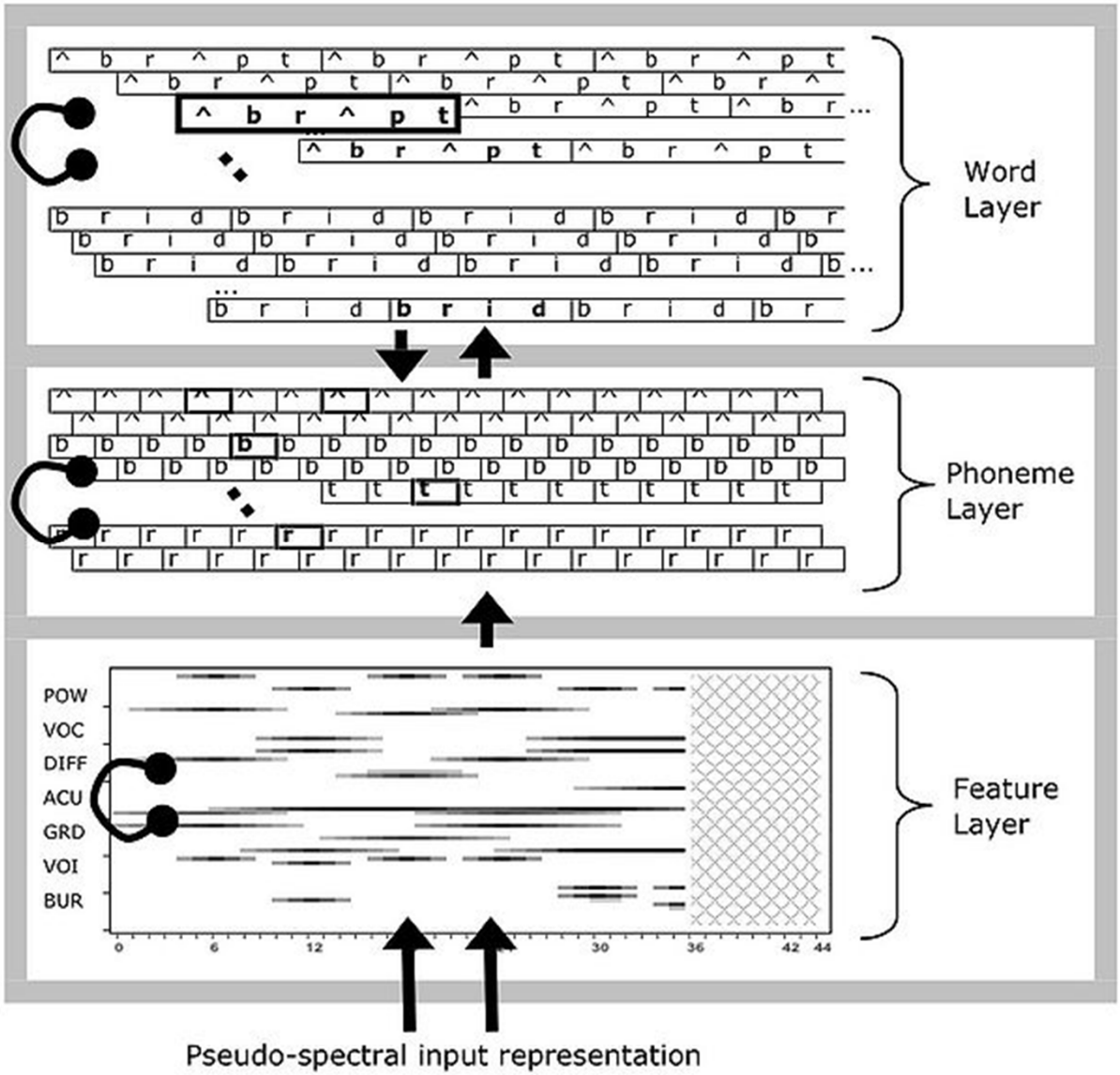


Fig. 6. Effect of VOT and time on fixations to the competitor.

This finding is consistent with the idea that activation flows in a *cascaded* way from features to phonemes.



Even stimuli that are identified as, say, /b/, 100% of the time can vary in how strongly and unambiguously the "voiced" feature is activated.

If the stimulus is obviously +bilabial and +stop, this creates some momentary ambiguity between /b/ and /p/.

That ambiguity is reflected in eye-movements while the participant decides where to click.

Evidence that partial activation can reach all the way to the lexical level

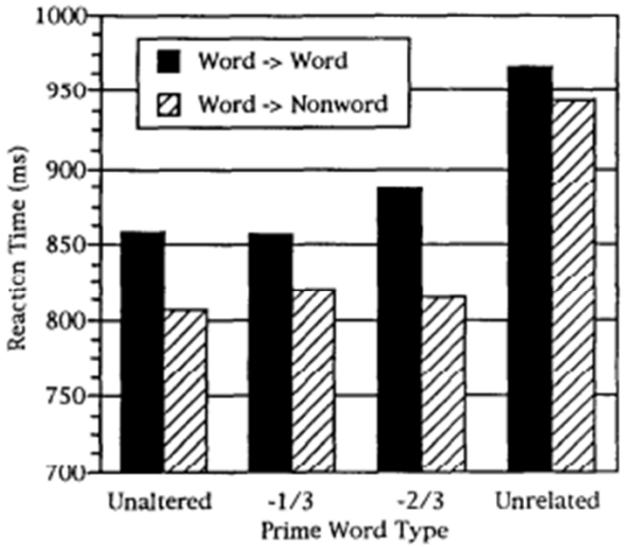


Figure 2. *Mean RTs to target words in the lexical decision task in Experiment 1. Black bars indicate targets preceded by primes whose voiced counterparts are real words ("pet → bet"). Shaded bars indicate targets preceded by primes with nonword voiced counterparts ("king → ging"). Unaltered, $-\frac{1}{3}$ and $-\frac{2}{3}$ files are semantically related to the target.*

Andruski et al. (1994) showed that you can even get semantic priming with nonwords.

So, "beace" primes "war," and "ging" primes "queen."

(the "word -> word" items are things like "bill" priming "drug," etc.)

Thus, it seems that lexical representations can be activated even by incomplete

That's one piece of evidence for *cascaded* processing in speech perception.

It would be hard to explain why the number of looks to the other member of the minimal pair increases gradually as you approach the category boundary in a thresholded model…

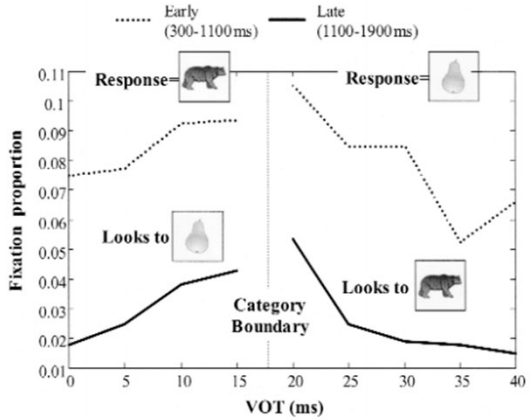
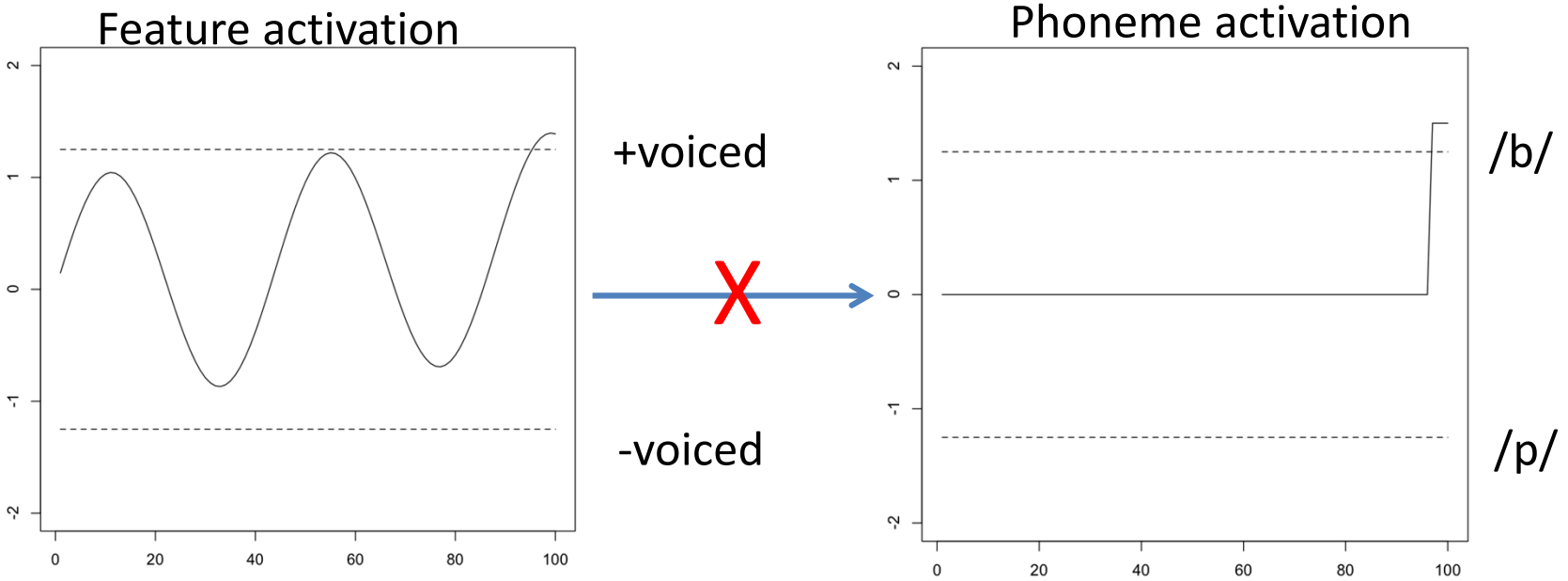
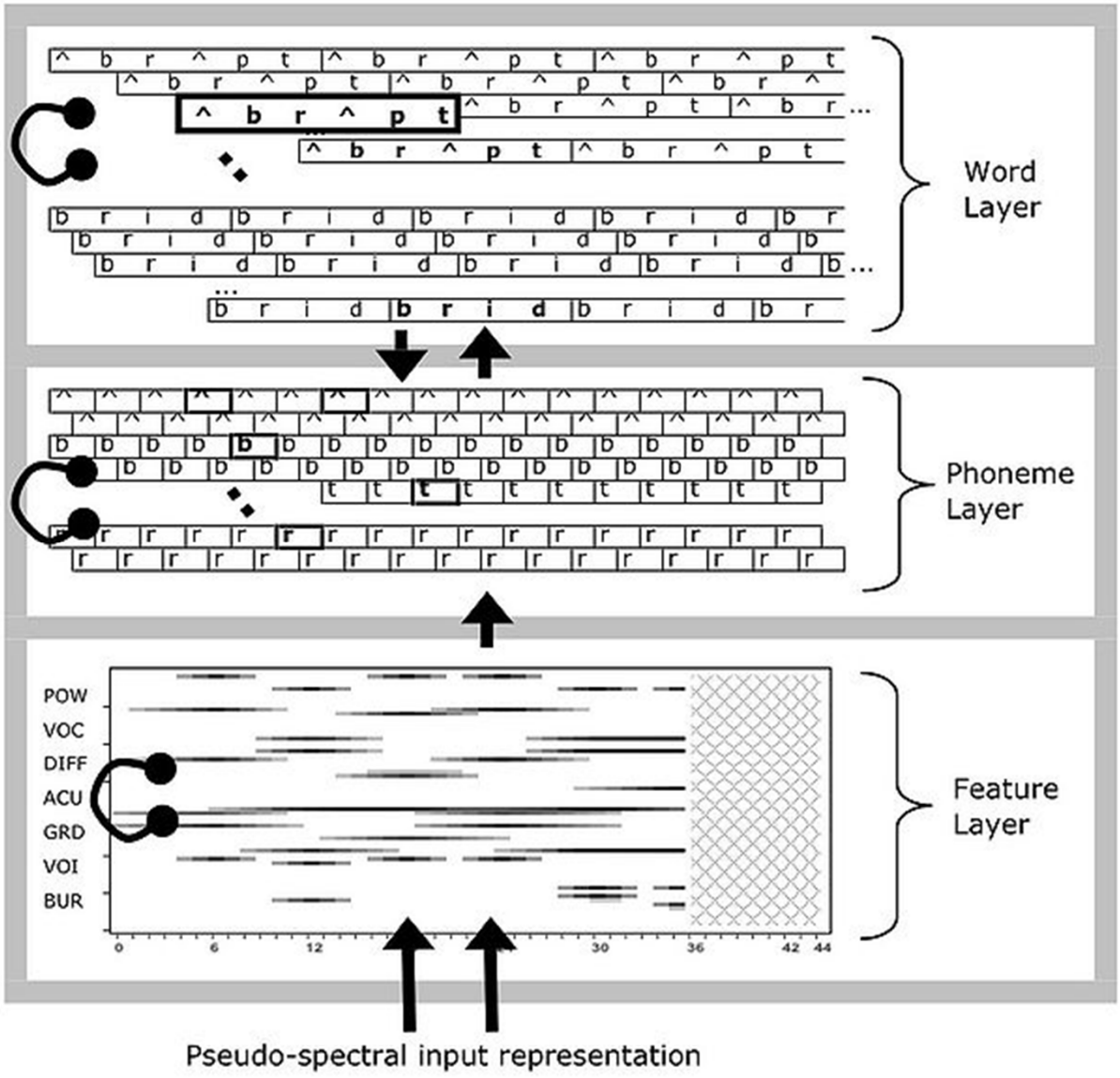


Fig. 6. Effect of VOT and time on fixations to the competitor.



Note:

This model also incorporates *feedback* from the lexical layer to the phoneme layer



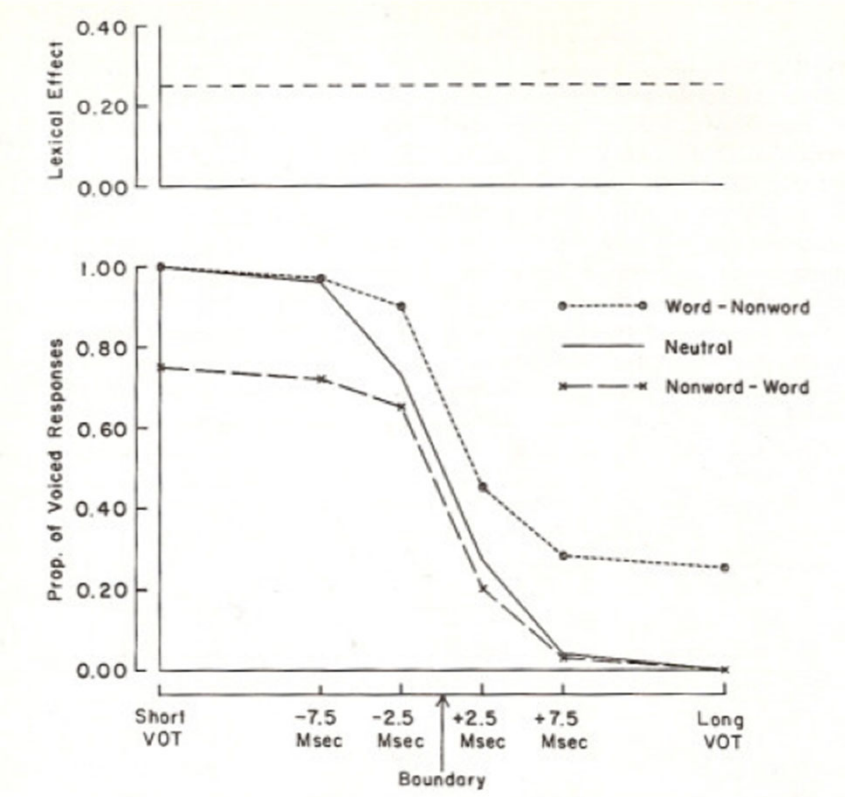
Next, we'll take a look at some experiments that motivate incorporating this kind of feedback.

Gisses and kifts for Valentine's day: The Ganong effect.

Ganong (1980) looked at how phoneme identification is impacted by word knowledge.

He came up with word/nonword pairs like "corpse/gorpse" and produced continua spanning the relevant phoneme boundary in 5ms steps.

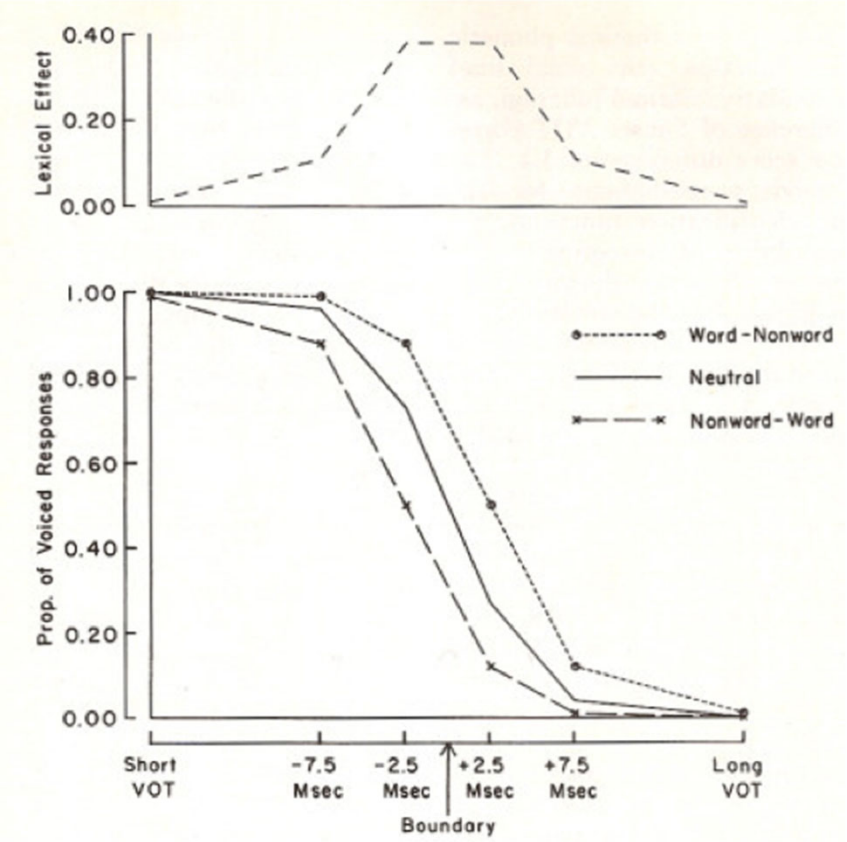
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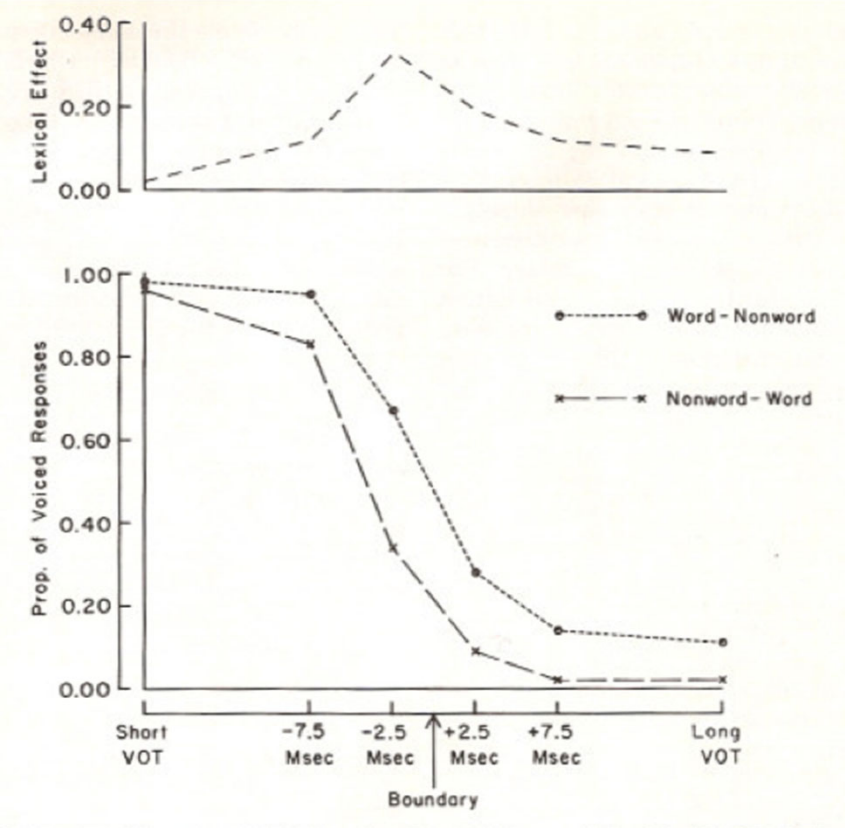
He came up with word/nonword pairs like "corpse/gorpse" and produced continua spanning the relevant phoneme boundary in 5ms steps.

Here's a prediction based on a late, "correction" process, whereby categorical perception functions as usual, but there is a bias to report having heard a word.



Here's a prediction of what would happen if word knowledge instead worked to *shift the categorical boundary.*

Note that the lexical effect is strongest for the items closest to the category boundary, because those are the stimuli for which lexical information can help resolve some ambiguity about the initial phoneme.



The results are pretty strikingly consistent with the second model.

Ganong raises a number of possibilities for how this could happen, focusing on different kinds of guessing strategies.

But later models would emphasize the potential role of feedback in these shifts.

A simulation of the Ganong effect in the TRACE model:

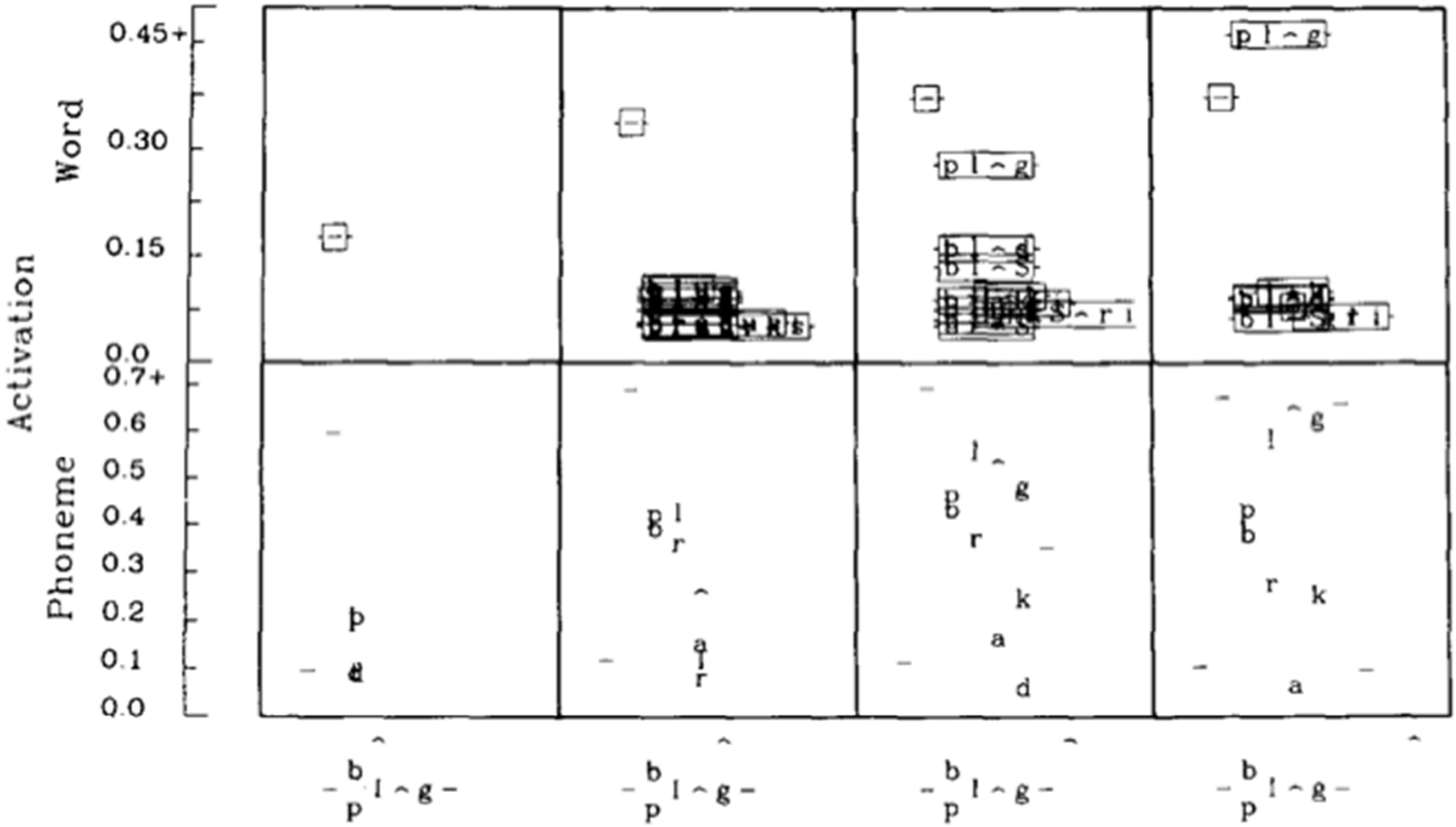


FIG. 5. Phoneme- and word-level activations at several points in the unfolding of a segment ambiguous between /b/ and /p/, followed by /l/, /ˆ/, and /g/. See text for a full explanation.

A simulation of the Ganong effect in the TRACE model:

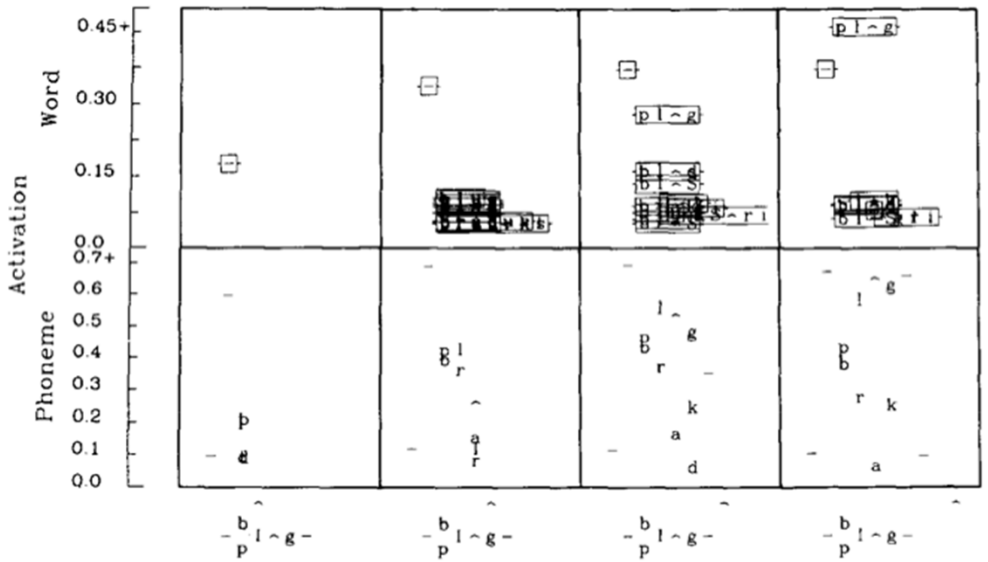
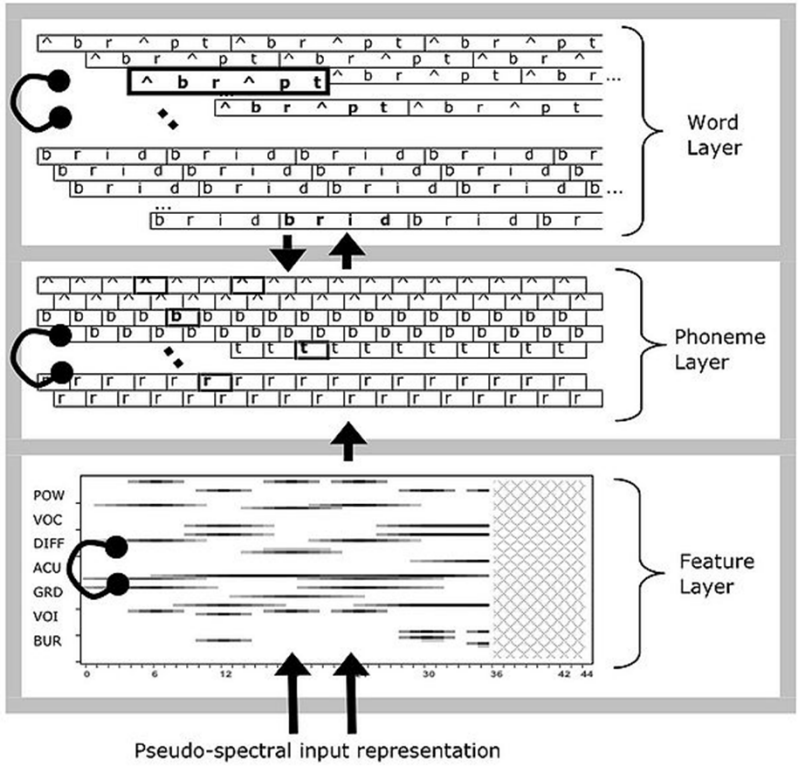


FIG. 5. Phoneme- and word-level activations at several points in the unfolding of a segment ambiguous between /b/ and /p/, followed by /l/, /ɹ/, and /g/. See text for a full explanation.



As the input unfolds over time, activity accumulates for “plug,” especially when the final /g/ arrives, which is highly inconsistent with “blood” and “blush” (also activated in Frame 3).

Thus, this model explains word effects on phoneme category boundaries in terms of *top-down* processing. Specifically, activity from the lexical level can produces *feedback* to the phoneme level, producing greater activation for the phoneme consistent with a word.

This should remind you of something from a previous lecture!

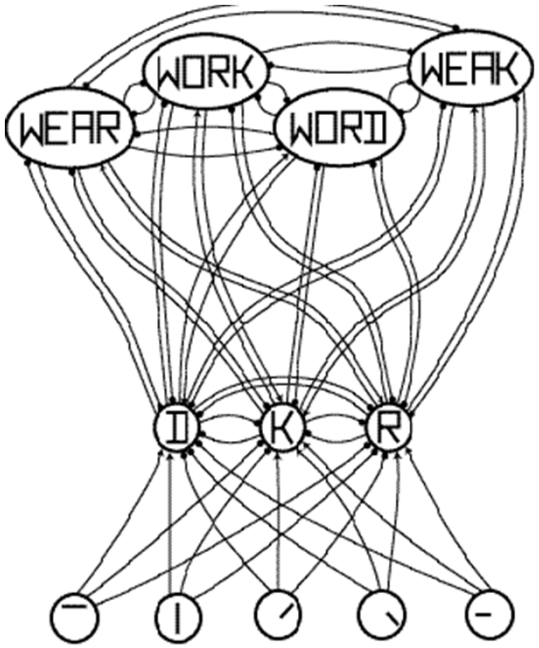


Figure 1. Interactive Activation Network Model (after McClelland and Rumelhart, 1981).



Figure 5. A hypothetical set of features that might be extracted on a trial in an experiment on word perception.

In fact the TRACE model is a version of the Interactive Activation model for written word recognition.

Many of the principles are the same, but a key issue in speech perception is the way the stimuli unfold over time.

Categorical perception and the “lexical-phonetic interface”

As we saw last week, perception appears to be shaped by category boundaries.

Specifically: Given the same physical difference, it is more difficult to distinguish stimuli from the same category than from different categories.

Nothing we learned today really undermines that basic finding.

BUT:

Ambiguity about phoneme identity *does* have an impact on word recognition.

Word-level information seems to have an impact on phoneme categorization.

Thus, it can’t be the case that the word recognition system just sits there waiting for the phoneme recognition system to finish doing its job (like a lumber mill). Spoken word recognition seems to be more *interactive* than this.

For next session:

Read:

Altmann, G. T. M. (1998) Ambiguity in Sentence Processing. *Trends in Cognitive Sciences, 2,* 146-151.

http://www.psycholinguistics.com/gerry_altmann/resources/TICS.pdf

Not entirely relevant, but the description of the food web reminded me of this documentary, which is pretty critical of classical notions of ecology, in ways that I (and maybe only I) find sort of relevant to cognitive science…

https://vimeo.com/groups/96331/videos/80799352

Spoken word recognition is also very *fast.*

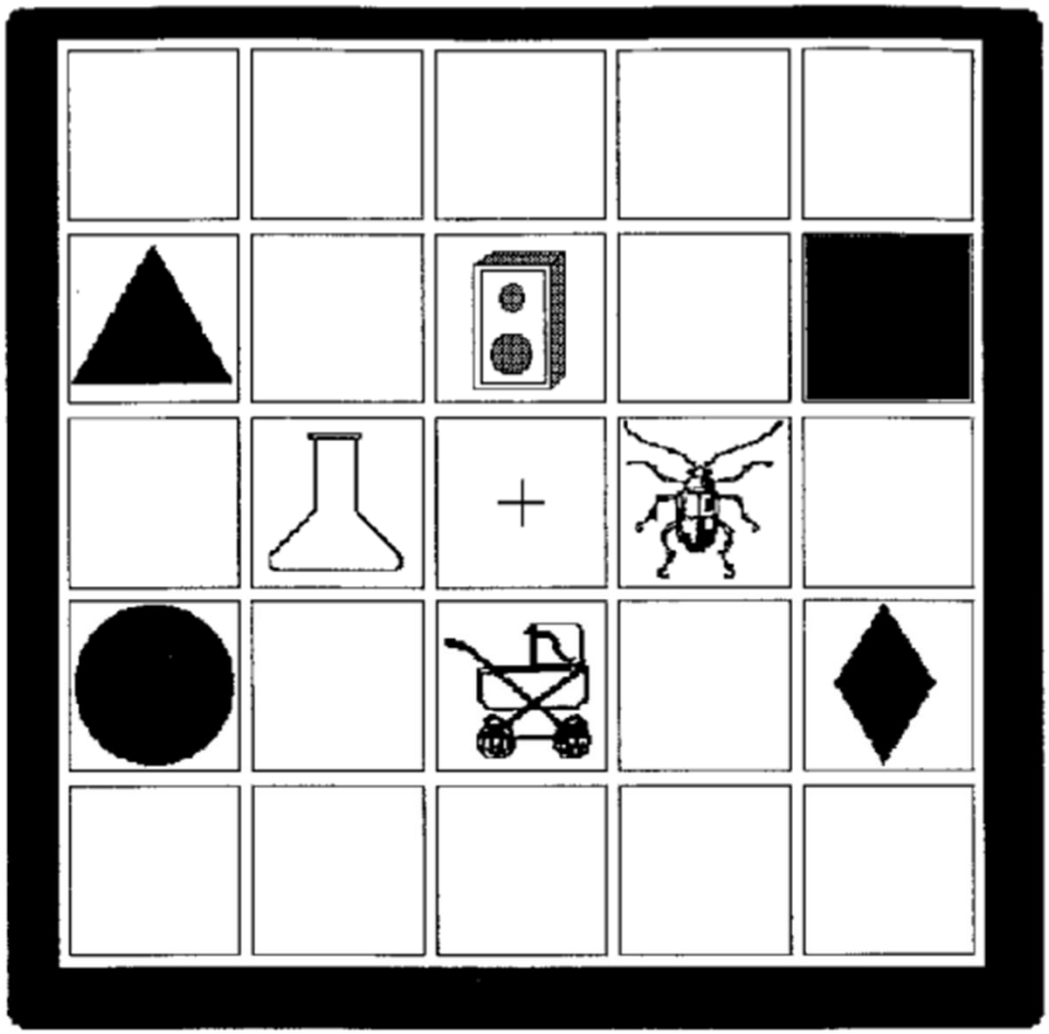


FIG. 3. An example of a stimulus display presented to participants.

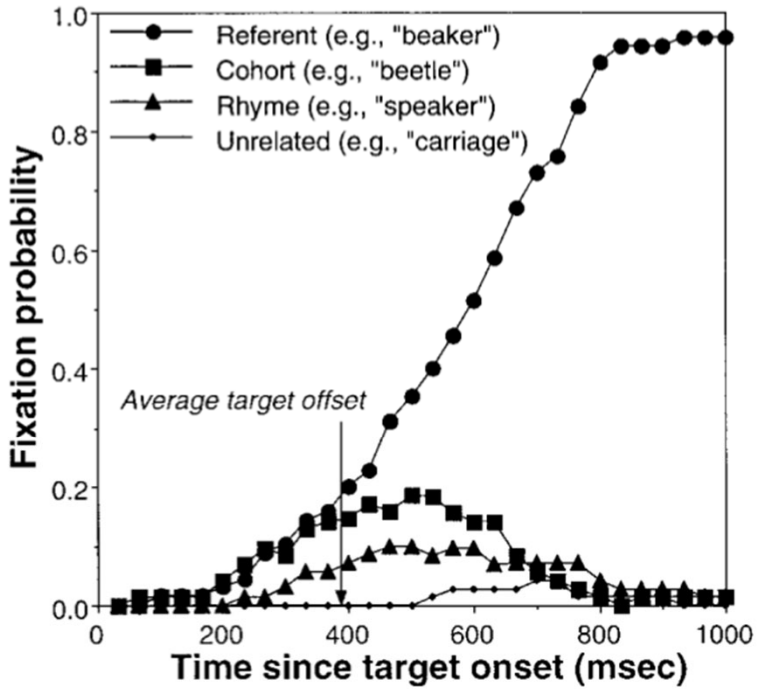
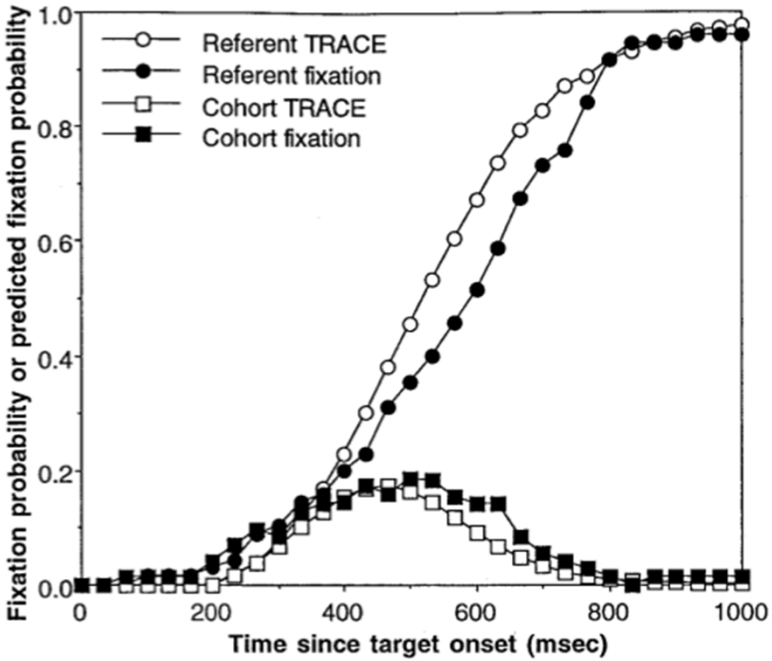
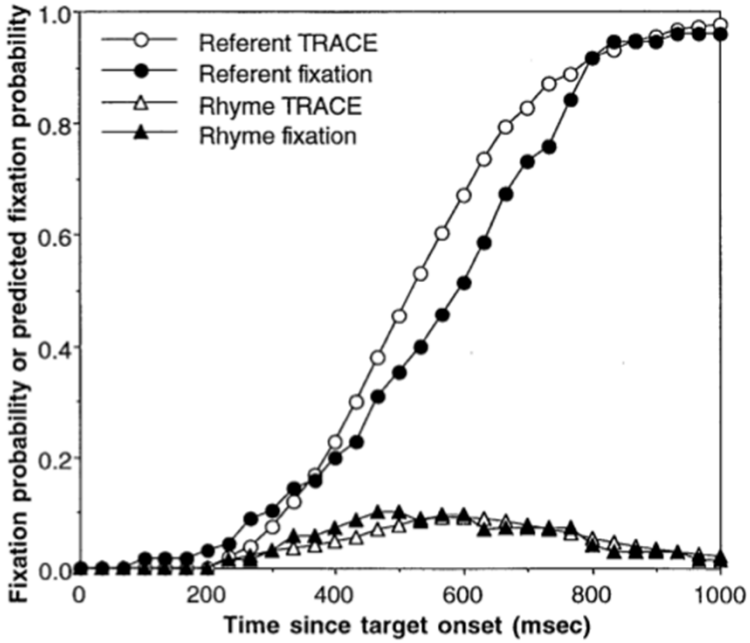


FIG. 4. Probability of fixating each item type over time in the full competitor condition in Experiment 1. The data are averaged over all stimulus sets given in Table 1; the words given in the figure are examples of one set.

Allopenna et al. (1994, JML)





Note the granularity of the simulation. Since eye-movements give us an idea about the changes in “activity” for different competitors over time, we can see cognitive processing before a final response is issued.

(Remember the graded effects of category goodness in the categorical perception task?)